PHASE 3: COMPLETE STUDY GUIDE

Work & Chain Rule + Series + Coding-Decoding | Target: 1 Hour | Pattern Recognition

TOPIC 1: WORK & CHAIN RULE (20 minutes)

1A. BASIC WORK CONCEPTS (3 min)

The Core Idea:

If A finishes a job in 'a' days, then in 1 day A does **1/a** of the work.

A's 1 day work = 1/a

When A and B work together:

Combined 1 day work = 1/a + 1/b

Time together = ab / (a + b)

Example 1:

A does work in 10 days, B in 15 days. Together?

Together = (10 x 15)/(10 + 15) = 150/25 = **6 days**

When one person leaves:

If A and B together take T days, B alone = aT / (a - T)

Example 2:

A and B together do work in 12 days. A alone in 20 days. B alone?

B = (20 x 12)/(20 - 12) = 240/8 = **30 days**

Efficiency-based problems:

If A is k times as efficient as B: A's time = B's time / k

Example 3:

A is 60% more efficient than B. A takes 12 days. B takes?

B's time = 12 x 1.6 = **19.2 days**. Or: A does 1.6 units/day. If B does 1 unit/day, B = 12 x 1.6 = 19.2 days.

Partial work problems:

Example 4:

A does work in 20 days. After 5 days, B joins. Together they finish in 3 more days. B alone?

A works 8 days total: 8/20 = 2/5. B works 3 days: 3/b. Total = 1.

3/b = 1 - 2/5 = 3/5. b = 3 x 5/3 = **5 days**

1B. CHAIN RULE (5 min - THE BIG FORMULA)

$$M_1 \times D_1 \times H_1 / W_1 = M_2 \times D_2 \times H_2 / W_2$$

M = Men (or workers), D = Days, H = Hours per day, W = Work done

How to remember which is direct/inverse:

More men => Less days needed (INVERSE)

More work => More days needed (DIRECT)

More hours/day => Less days needed (INVERSE)

Example 5:

12 men, 20 days, 8 hrs/day to finish a work. 15 men, 6 hrs/day => how many days?

12 x 20 x 8 / 1 = 15 x D x 6 / 1

1920 = 90D => D = **21.33 days**

Example 6:

5 men complete work in 10 days working 6 hrs/day. How many men needed to complete double the work in 8 days working 5 hrs/day?

5 x 10 x 6 / 1 = M x 8 x 5 / 2. Note: W2 = 2 (double work).

300 = 40M/2 = 20M. M = **15 men**

1C. PIPES & CISTERNS (3 min)

Same concept as work! Inlet = positive work, Outlet = negative work.

Inlet fills in 'a' hours: Rate = +1/a

Outlet empties in 'b' hours: Rate = -1/b

Both open: Net rate = 1/a - 1/b | Time to fill = ab/(b-a) [when b > a]

Example 7:

Pipe A fills in 6 hours. Pipe B empties in 10 hours. Both open, time to fill?

Net rate = 1/6 - 1/10 = (10-6)/60 = 4/60 = 1/15. Time = **15 hours**

Example 8:

Pipe A fills in 4 hrs, B fills in 6 hrs, C empties in 12 hrs. All open?

Net = 1/4 + 1/6 - 1/12 = 3/12 + 2/12 - 1/12 = 4/12 = 1/3. Time = **3 hours**

WORK & CHAIN RULE: PRACTICE QUESTIONS

W1. A does work in 10 days, B in 15 days. Together?

a) 5 days
b) 6 days
c) 7 days
d) 8 days

W2. 12 men, 20 days to complete. 15 men take how many days?

a) 14
b) 16
c) 18
d) 25

W3. A is twice as fast as B. Together they finish in 12 days. A alone?

a) 15 days
b) 18 days
c) 20 days
d) 24 days

W4. Pipe A fills in 8 hrs, B empties in 12 hrs. Both open, fill time?

a) 20 hrs
b) 24 hrs
c) 10 hrs
d) 16 hrs

W5. 6 men and 8 women finish work in 10 days. 26 men and 48 women finish in 2 days. 15 men and 20 women?

a) 3 days
b) 4 days
c) 5 days
d) 6 days

WORK & CHAIN RULE: SOLUTIONS

W1. Answer: b) 6 days | ab/(a+b) = 150/25 = 6 days.

W2. Answer: b) 16 | M1xD1 = M2xD2. 12x20 = 15xD. D = 240/15 = 16.

W3. Answer: b) 18 days | Let B's rate = x, A's = 2x. Together: 3x = 1/12, x = 1/36. A = 2/36 = 1/18. A alone = 18 days.

W4. Answer: b) 24 hrs | Net = 1/8 - 1/12 = (3-2)/24 = 1/24. Time = 24 hrs.

W5. Answer: b) 4 days | Let 1 man = m, 1 woman = w per day. (6m+8w)10=1 and (26m+48w)2=1. So 60m+80w=1 and 52m+96w=1. Solving: 8m=16w => m=2w. Put back: 60(2w)+80w=200w=1 => w=1/200. m=1/100. (15/100+20/200)D=1 => (0.15+0.1)D=1 => D=4.

TOPIC 2: SERIES - NUMERICAL & ALPHABET (20 minutes)

2A. THE 3-STEP METHOD FOR ANY NUMBER SERIES (3 min)

Step 1: Write the differences between consecutive terms.

Step 2: If differences aren't constant, write differences of differences.

Step 3: Identify the pattern (constant diff, AP, GP, squares, cubes, primes, etc.)

2B. COMMON PATTERNS WITH EXAMPLES (10 min - STUDY ALL)

Pattern 1: CONSTANT DIFFERENCE (Arithmetic Progression)

3, 7, 11, 15, 19, ? => Diff: +4, +4, +4, +4 => Next = **23**

Pattern 2: CONSTANT RATIO (Geometric Progression)

2, 6, 18, 54, ? => Ratio: x3, x3, x3 => Next = **162**

Pattern 3: INCREASING DIFFERENCE (AP in differences)

2, 6, 12, 20, 30, ? => Diff: 4, 6, 8, 10 (AP, +2) => Next diff = 12 => **42**

This is n(n+1) series: 1x2=2, 2x3=6, 3x4=12, 4x5=20, 5x6=30, 6x7=42

Pattern 4: SQUARE SERIES

1, 4, 9, 16, 25, 36, ? => These are 1^2, 2^2, 3^2, ... => **49**

Variations: n^2+1: 2,5,10,17,26,? => **37** | n^2-1: 0,3,8,15,24,? => **35**

Pattern 5: CUBE SERIES

1, 8, 27, 64, 125, ? => These are 1^3, 2^3, 3^3, ... => **216**

Pattern 6: FIBONACCI-LIKE (each = sum of previous two)

1, 1, 2, 3, 5, 8, 13, ? => 8+13 = **21**

Variant: 2, 3, 5, 8, 13, 21, ? => Same pattern, different start => **34**

Pattern 7: ALTERNATING OPERATIONS

2, 4, 12, 14, 42, 44, ? => Pattern: +2, x3, +2, x3, +2, x3 => **132**

Pattern 8: PRIME NUMBER ADDITIONS

3, 5, 8, 13, 21, ? => Diff: 2, 3, 5, 8... hmm, not primes.

Better example: 1, 3, 6, 11, 18, 29, ? => Diff: 2, 3, 5, 7, 11 (PRIMES!) => Next prime = 13 => **42**

Pattern 9: MULTIPLY & ADD/SUBTRACT

3, 7, 15, 31, 63, ? => Each = prev x 2 + 1 => 63 x 2 + 1 = **127**

Also: these are 2^2-1, 2^3-1, 2^4-1, 2^5-1, 2^6-1, 2^7-1 = 127

Pattern 10: DOUBLE DIFFERENCES

3, 8, 18, 35, 61, ?

1st diff: 5, 10, 17, 26 **|** 2nd diff: 5, 7, 9 (AP, +2) => Next 2nd diff = 11

Next 1st diff = 26 + 11 = 37 **|** Answer = 61 + 37 = **98**

2C. ALPHABET SERIES (5 min)

Step 1: MEMORIZE alphabet positions:

A=1	B=2	C=3	D=4	E=5	F=6	G=7	H=8	I=9
J=10	K=11	L=12	M=13	N=14	O=15	P=16	Q=17	R=18
S=19	T=20	U=21	V=22	W=23	X=24	Y=25	Z=26	

Opposite pairs (sum = 27): *A-Z, B-Y, C-X, D-W, E-V, F-U, G-T, H-S, I-R, J-Q, K-P, L-O, M-N*

Common Alphabet Patterns:

Skip pattern: B, D, G, K, P, ? => Gaps: +2, +3, +4, +5, +6 => P(16)+6 = V(22) => **V**

Constant skip: A, D, G, J, M, ? => Gap: +3 each => M(13)+3 = P(16) => **P**

Reverse: Z, W, T, Q, N, ? => Gap: -3 each => N(14)-3 = K(11) => **K**

Two interleaved: A, Z, C, X, E, V, ? => Odds: A,C,E,G(+2) **|** Evens: Z,X,V,T(-2) => **G, T**
Group pattern: AB, DE, GH, ? => Skip one letter between groups => **JK**

SERIES: PRACTICE QUESTIONS

S1. Find next: 5, 11, 23, 47, 95, ?

a) 189
b) 191
c) 185
d) 193

S2. Find next: 1, 4, 13, 40, 121, ?

a) 364
b) 360
c) 363
d) 365

S3. Find next: 2, 3, 5, 8, 13, 21, ?

a) 29
b) 32
c) 34
d) 36

S4. Find next: B, D, G, K, P, ?

a) T
b) U
c) V
d) W

S5. Find next: Z, X, U, Q, L, ?

a) F
b) G
c) H
d) E

S6. Find the missing: 1, 4, 27, 256, ?

a) 3025
b) 3125
c) 3200
d) 2500

SERIES: SOLUTIONS

S1. Answer: b) 191 | Each = prev x 2 + 1. 95 x 2 + 1 = 191.

S2. Answer: a) 364 | Each = prev x 3 + 1. 121 x 3 + 1 = 364.

S3. Answer: c) 34 | Fibonacci-like: each = sum of previous two. 13 + 21 = 34.

S4. Answer: c) V | Gaps: +2, +3, +4, +5, +6. P(16)+6 = V(22).

S5. Answer: a) F | Gaps: -2, -3, -4, -5, -6. L(12)-6 = F(6).

S6. Answer: b) 3125 | Pattern: 1^1, 2^2, 3^3, 4^4, 5^5. 5^5 = 3125.

TOPIC 3: CODING - DECODING (20 minutes)

3A. THE 5 TYPES OF CODES (15 min)

TYPE 1: CONSTANT SHIFT (Most Common!)

Each letter shifts by the SAME number of positions.

How to detect: Subtract positions of coded letters from original. If constant, it's a shift.

Example 1: COMPUTER => DPNQVUFS

C(3)->D(4), O(15)->P(16), M(13)->N(14), P(16)->Q(17)...
Each letter shifted by +1. So PRINTER => **QSJOUFS**

Example 2: CAT => FDW

C(3)->F(6), A(1)->D(4), T(20)->W(23). Shift = +3 each.
So DOG => **GRJ**

TYPE 2: INCREASING/VARIABLE SHIFT

Each letter shifts by a different amount (often +1, +2, +3... or +2, +3, +4...)

Example 3: FRIEND => HUMJTK

F(6)+2=H(8), R(18)+3=U(21), I(9)+4=M(13), E(5)+5=J(10), N(14)+6=T(20), D(4)+7=K(11)
Pattern: +2, +3, +4, +5, +6, +7 (increasing shift)
So CANDLE: C+2=E, A+3=D, N+4=R, D+5=I, L+6=R, E+7=L => **EDRIRL**

TYPE 3: REVERSAL

The word is simply written backwards.

Example 4: TIGER => REGIT

So HORSE => **ESROH**

TYPE 4: MIRROR/OPPOSITE CODING

Each letter replaced by its opposite: A<->Z, B<->Y, C<->X ... (sum = 27)

Coded letter = 27 - Original position

A-Z	**B-Y**	**C-X**	**D-W**	**E-V**	**F-U**	**G-T**
H-S	**I-R**	**J-Q**	**K-P**	**L-O**	**M-N**	

Example 5: HELLO => SVOOL

H(8)->S(19), E(5)->V(22), L(12)->O(15), L(12)->O(15), O(15)->L(12)
So WORLD: W->D, O->L, R->I, L->O, D->W => **DLIOW**

TYPE 5: NUMBER/SYMBOL SUBSTITUTION

Letters replaced by their position values or by symbols per a given code table.

Example 6: If A=1, B=2, ... then FACE = ?

F=6, A=1, C=3, E=5. Code = **6-1-3-5**

3B. THE SOLVING APPROACH (2 min read)

Step 1: Write down the position number of each letter in BOTH original and coded words.

Step 2: Find the difference between corresponding positions.

Step 3: Check if difference is constant, increasing, or follows a pattern.

Step 4: If none work, check for reversal or mirror coding.

Step 5: Apply the SAME rule to the new word.

Pro tip: *In exams, always start by checking +1 or +2 shift first. It covers >50% of questions.*

CODING-DECODING: PRACTICE QUESTIONS

CD1. If TIGER = UJHFS, then HORSE = ?

a) IPSFT
b) IPSTF
c) GNSQD
d) JQTUF

CD2. If PAPER = SDSHU, then COVER = ?

a) FRYHU
b) FRYLU
c) FRYHU
d) FRXHU

CD3. If GREAT = TIVZG, then BUILD = ?

a) YFROW
b) YFROW
c) YFROD
d) YFROC

CD4. If SIT = VLW, then STAND = ?

a) VWDQG
b) VXDQG
c) VWCPF
d) VWDQH

CD5. If ROSE = TQUG, what is the pattern and code for LILY?

a) NKNA (+2 shift)
b) NJNB (+2,-1 alternating)
c) NKNA
d) NJMA

CODING-DECODING: SOLUTIONS

CD1. Answer: b) IPSTF | Shift +1: T->U, I->J, G->H, E->F, R->S. So H->I, O->P, R->S, S->T, E->F = IPSTF.

CD2. Answer: a) FRYHU | Shift +3: P->S, A->D, P->S, E->H, R->U. So C->F, O->R, V->Y, E->H, R->U = FRYHU.

CD3. Answer: a) YFROW | Mirror code (27-pos): G->T, R->I, E->V, A->Z, T->G. So B->Y, U->F, I->R, L->O, D->W = YFROW.

CD4. Answer: a) VWDQG | S+3=V, I+3=L, T+3=W. Shift +3. S+3=V, T+3=W, A+3=D, N+3=Q, D+3=G = VWDQG.

CD5. Answer: a) NKNA (+2 shift) | R+2=T, O+2=Q, S+2=U, E+2=G. All +2. L+2=N, I+2=K, L+2=N, Y+2=A(wraps). = NKNA.

PHASE 3: QUICK REVISION CARD

WORK & CHAIN RULE

1. A's 1 day work = 1/a | Together = ab/(a+b)
2. Chain Rule: M1.D1.H1/W1 = M2.D2.H2/W2
3. Pipes: Inlet = +1/a, Outlet = -1/b, Both = ab/(b-a)

SERIES

1. Always find differences first, then differences of differences
2. Common: AP, GP, squares, cubes, Fibonacci, alternating ops, prime additions
3. Alphabet: convert to numbers, find gaps, apply pattern
4. Opposites sum to 27: A-Z, B-Y, C-X, ..., M-N

CODING-DECODING

1. Constant shift: same +k to each letter (check +1, +2, +3 first)
2. Variable shift: +1,+2,+3... or +2,+3,+4...
3. Mirror code: A<->Z, B<->Y (pos + coded pos = 27)
4. Reversal: word written backwards
5. APPROACH: Write positions -> Find diffs -> Identify pattern -> Apply

ALL 3 PHASES COMPLETE! Now solve the MCQ practice papers, then review cheat sheet before sleeping.

Good luck on your exam, Angelina!